

Playwright

Modern Web automation for E2E, component and API testing across Chromium, Firefox and WebKit.

What it is and when to use it

Playwright is a modern Web automation framework suited to E2E, regression, smoke and cross-browser flows, with isolated contexts, parallel execution and rich diagnostics.

Prerequisites

For JavaScript/TypeScript, install a supported Node.js version and package manager. Verify `node --version` and `npm --version`. Git and VS Code are recommended.

Step-by-step installation

Run `npm init playwright@latest` in a new or existing project. The wizard creates the initial structure and can install browsers. Existing projects can install `@playwright/test` and run `npx playwright install`.

Initial configuration

Review `playwright.config` for `baseURL`, `projects`, `retries`, `workers`, `reporters`, `screenshots`, `video` and `traces`. Keep secrets and environment URLs in environment variables.

First test

Create a `.spec.ts` file, navigate with `page.goto`, prefer `getByRole/getByLabel` and assert with `expect`. Run `npx playwright test` and inspect the HTML report.

Locators and synchronization

Prefer semantic locators and web-first assertions. Avoid fixed sleeps; use Playwright auto-waiting and explicit state expectations.

Debug and evidence

Use UI Mode, Inspector, Trace Viewer, screenshots and video to identify product, environment, data or automation failures.

CI/CD

Install dependencies and browsers in CI, run the suite, and publish reports/traces/screenshots as artifacts.

Best practices and troubleshooting

Keep tests independent, use resilient locators, isolate data, control authentication state and investigate flaky tests rather than hiding them.

QA tip

Use the tool within a quality strategy: automate meaningful risks, keep the suite observable and treat automation failures as production-code issues.

Official resources

Site oficial: <https://playwright.dev/>

Download / instalação: <https://playwright.dev/docs/intro>

Documentação: <https://playwright.dev/docs/intro>